Thermodynamically and Spectroscopically Informed Optical Chemical Structure Recognition and the ChemVeritas-600 Multimodal Science Benchmark

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Abstract

Automated mining of organic synthesis and natural product Supporting Information (SI) is crippled by *silent stereochemical corruption*: when bridged polycycles, contiguous stereocenters, or low-DPI scans obscure wedge/dash geometry, vision-only Optical Chemical Structure Recognition (OCSR) emits valid-valence SMILES whose two-dimensional connectivity is correct yet whose three-dimensional configuration encodes inverted diastereomers or severely strained ring junctions. Manual validation of million-scale SI extractions (`NMRexp`) reveals a 17% stereochemical corruption rate even on single-stereocenter molecules, while recent 2D graph-verification pipelines (`TRACE-Chem`) remain blind to diastereomers sharing identical formulas and topological graphs. Here we formulate structure recognition as closed-loop *joint visual–thermodynamic–spectroscopic Bayesian inference*, pairing planar C_8-equivariant visual beam decoding (`Ouroboros`) and targeted 2^m uncertain-stereocenter epimer enumeration with continuous 3D `MACE-OFF23-medium` Boltzmann strain energetics, exact-mass HRMS filtering, $O(3)$ local-Jacobian symmetry-orbit ^{13}C peak counting, unassigned Hungarian ^{13}C/^{1}H chemical-shift matching, and Karplus $^{3}J_{\mathrm{HH}}$ dihedral constraints. To benchmark both OCSR pipelines and frontier multimodal foundation models, we introduce **`ChemVeritas-600`**, spanning 600 connectivity-disjoint polycyclic, near-Bredt, superatom, and modern 3D bioisostere scaffolds across five strictly segregated visual views (2,400 visual specifications) and five reasoning tracks (`T1`–`T5`) with four-step Reasoning Path Fidelity (`Veritas-RPF`). Coupled to real pretrained `Ouroboros-AC` weights ($n = 100$ per view), Bayesian physical–spectroscopic inference lifts exact stereoisomer recovery from 1.0% [0.0, 3.0] to 14.0% [7.0, 21.0] on canonical clean depictions (`V0`), from 1.0% [0.0, 3.0] to 12.0% [6.0, 19.0] on off-grid rotated depictions (`V1`), from 1.0% [0.0, 3.0] to 3.0% [0.0, 7.0] on crowded SI scans (`V2`), and from 0.0% [0.0, 0.0] to 4.0% [1.0, 8.0] on condensed superatom layouts (`V3`). Across $N = 50$ silent stereochemical corruptions, `MACE-OFF23-medium` BFGS relaxation exposes a median strain penalty of +45.00 kcal/mol (ROC-AUC = 0.853). On `ChemVeritas-600`, full posterior inference reaches 77.0% [73.8, 80.3] `T2` stereochemical accuracy (93.2% `Veritas-RPF`) and 100.0% calibrated abstention when achiral observables cannot resolve enantiomers. Finally, zero-retraining evaluation on $N = 30$ authentic published experimental ^{13}C/^{1}H/HRMS records (15 literature isomer pairs; ^{13}C MAE = 5.25 ppm, $r = 0.9810$) recovers 73.3% [56.7, 86.7] (`MMFF94s`) and 66.7% [50.0, 83.3] (`MACE-OFF23-medium//MMFF94s`) of true stereoisomers from adversarially down-weighted beams.

1 Introduction

Chemical structure depictions are the primary visual language through which synthetic organic, natural product, and medicinal chemists communicate molecular identity [1–3]. Over the past five years, deep-learning Optical Chemical Structure Recognition (OCSR) architectures—spanning image-to-SELFIES/SMILES transformers (`DECIMER.ai` [1], `SwinOCSR` [4], `ChemPix` [5]), conditional image-to-graph decoders (`MolScribe` [2], `MolNexTR` [6]), keypoint-and-GNN detectors (`MolGrapher` [7], `AtomLenz` [8]), and reinforcement-learning vision-language models (`MolSight` [9])—have pushed two-dimensional (2D) constitutional skeleton accuracy above 95% on clean patent renderings. Yet as automated extraction pipelines scale to hundreds of thousands of real Supporting Information (SI) PDFs [10–13], a pervasive failure mode has emerged: *silent stereochemical corruption*.

Unlike syntax errors that fail RDKit valence sanitization, a silent stereochemical corruption is a valid-valence, canonically parsable molecule whose 2D constitution matches the depicted compound (or drops a single H/C under superatom compression), but whose tetrahedral (`@`/`@@`) or bridged-ring configuration has been inverted by compressed wedge/dash pixels, crossing cage bonds, perspective chair/Haworth projections, or arbitrary planar diagram rotations. The empirical magnitude of this crisis was recently exposed by `NMRexp` [12], which applied `MolScribe` across ∼200,000 open-access SI docu-

ments to assemble 3.3 million experimental NMR records: manual technical validation confirmed 98% accuracy on flat 2D skeletons, but stereochemical accuracy dropped to 83% *even for single-stereocenter molecules*—injecting hundreds of thousands of inverted stereoisomers into downstream machine-learning training corpora [13, 14]. Even `MolSight` [9], which curated 200,000 confusing stereoisomer depictions (`Stereo-200K`), operates strictly from 2D pixels; when low-DPI rasterization blurs a dashed bond at a congested bridgehead, the true 3D configuration becomes *information-theoretically underdetermined from the image alone*.

Crucially, organic synthesis SIs almost never report a molecular depiction in isolation. Immediately adjacent to compound **3a** appears an experimental characterization block reporting high-resolution mass spectrometry (HRMS m/z within ± 5 ppm) and unassigned one-dimensional ^{13}C and ^{1}H NMR peak tables (δ in ppm, multiplicity, and vicinal coupling constants $^{3}J_{\mathrm{HH}}$ in Hz). In computational organic chemistry, Bayesian NMR shift and coupling analysis—pioneered by Goodman's `DP4` [15] and `DP4-AI` [16], Sarotti's `DP4+` [17] and `ML-J-DP4` [18], and accelerated by 3D neural/kernel NMR predictors (`CASCADE` [19], `IMPRESSION` [20]) alongside 2D GNNs (`SGNN` [21]) and multimodal spectral transformers [22–25]—routinely resolves diastereomers from a single experimental spectrum via Grant–Cheney γ-gauche steric shielding [26] and Karplus dihedral relationships [27]. Meanwhile, in 3D structural bioinformatics and physical organic chemistry, `PoseBusters` [28], `GenBench3D` [29], `CSD Mogul` [30], and transferable machine-learning interatomic potentials (`Auto3D` [31], `MACE-OFF23` [32]) have shown that valid 2D graphs frequently conceal severe 3D nonbonded clashes or ring strain, while McDermott, Houk, and Garg [33] overturned binary textbook Bredt's-rule heuristics by demonstrating that *continuous 3D strain energy* governs room-temperature isolability.

Despite this convergence, visual OCSR, 3D thermodynamic strain, and 3D spectroscopic elucidation have remained disconnected silos. The closest recent effort, `TRACE-Chem` [34], constructs a 2D evidence graph checking IUPAC names, `MolScribe` top-1 outputs, molecular formulas, and 2D GNN NMR shifts across 1,852 SI records. However, `TRACE-Chem` exhibits four fundamental limitations: (i) it incorporates zero 3D conformer generation or MLIP strain physics; (ii) when an SI refers to a structure only by compound label (**3a**) without a full $(1R, 2S, 4R)$ IUPAC name, all diastereomers share the exact same formula, exact same HRMS m/z, and exact same 2D topological graph, leaving 2D GNN verifiers stereochemically blind; (iii) it masks conflicting fields on top-1 outputs rather than enumerating and reranking uncertain stereocenters; and (iv) it provides no standardized benchmark for frontier multimodal reasoning models. Concurrently, existing chemistry LLM/VLM benchmarks—`ChemBench` [35], `MolPuzzle` [36], `LAB-Bench` [37], `ChemVLM` [38], `IsoBench` [39], `MMMU` [40], `SUPERChem` [41], `Humanity's Last Exam` [42], `FrontierScience` [43], `Mol-Instructions` [44], `LlaSMol` [45], and `ChemDFM-X` [46] (which aligns five modalities yet omits NMR entirely)—either test textbook *de novo* elucidation of simple achiral molecules without images or test isolated 2D OCSR without 3D thermodynamics or SI spectra.

Here we bridge these domains through five contributions:

1. **Group-Theoretic Visual-to-3D Symmetry Formulation:** We formalize the $SO(2)$ vs. $O(2)$ planar chirality boundary [47] alongside $SE(3)$ vs. $O(3)$ 3D scalar invariance, pairing C_8-steerable visual beam decoding with parity-invariant 3D conformer embedding and local $O(3)$ Jacobian determinant ($\det J_p = \pm 1$) symmetry-orbit partitioning.

2. **Closed-Loop Thermodynamic & Spectroscopic Posterior (`A0`–`A6`):** We combine continuous `MACE-OFF23-medium` [32] 3D strain and element-wise node-energy outlier penalties, exact-mass HRMS gates, 3D Boltzmann ^{13}C/^{1}H shift matching via Hungarian bipartite assignment, Karplus $^{3}J_{\mathrm{HH}}$ coupling constraints, and targeted 2^m ($m \leq 3$) uncertain-wedge epimer enumeration.

3. **`ChemVeritas-600` Frontier Benchmark:** We release a scaffold-disjoint benchmark of 600 complex organic scaffolds across five chemical strata (`S1`–`S5`), five strictly segregated visual views, five tasks (`T1`–`T5`), and four-step Reasoning Path Fidelity (`Veritas-RPF`).

4. **Closed-Loop Neural OCSR & `MACE-OFF23` Strain Validation:** Using real pretrained `Ouroboros-A/B/C/AC` weights across 500 multi-view images and $N = 50$ BFGS-relaxed strain pairs, we quantify both planar C_8 rotation invariance and dramatic stereochemical recovery on degraded scans.

5. **Authentic Literature NMR Validation:** Without fitting any parameters on the test set, we validate end-to-end isomer recovery on $N = 30$ authentic published experimental ^{13}C/^{1}H/HRMS records (15 classic terpene, norbornane, and cyclitol/cyclohexanol isomer pairs [48–52]).

2 Theory and Methodology

2.1 Planar $SO(2)$ vs. Spatial $O(3)$ Chirality Symmetries

A 2D chemical drawing $I : \mathbb{R}^2 \to [0, 1]$ is an orthographic projection with discrete wedge (up, $+z$) and dash (down, $-z$) depth annotations. Under an in-plane proper rotation $R_\theta \in SO(2)$, the projected (x, y) bond vectors rotate covariantly while $+z$ wedges remain $+z$

wedges; thus the represented 3D molecule M is strictly $SO(2)$-invariant [47]. Conversely, an in-plane reflection $s \in O(2) \setminus SO(2)$ reverses the signed 2D determinant of substituents at every tetrahedral center while leaving wedge/dash ink unchanged, mapping $M \mapsto \mathrm{mirror}(M)$ (its enantiomer). Conventional ResNet encoders lack $SO(2)$ equivariance and collapse under 90° diagram rotations, whereas $O(2)$-reflection-invariant pooling destroys Cahn–Ingold–Prelog (CIP) handedness. Our visual frontend (`Ouroboros-C` and ensemble `Ouroboros-AC`) therefore restricts steerable convolutions to the cyclic subgroup $C_8 \subset SO(2)$, preserving chiral handedness while achieving discrete rotation invariance.

A complementary symmetry governs the 3D physical and spectroscopic verifiers. In isotropic achiral media (gas phase or $CDCl_3$), the potential energy $E(\mathbf{R})$ and isotropic nuclear magnetic shieldings $\sigma_i(\mathbf{R})$ are invariant under the *full* orthogonal group $O(3) = SO(3) \times \mathbb{Z}_2$, including improper reflections $\mathbf{R} \mapsto -\mathbf{R}$:

$$E(M) = E(\mathrm{mirror}(M)), \qquad \mathcal{S}_{\mathrm{NMR}}(M) = \mathcal{S}_{\mathrm{NMR}}(\mathrm{mirror}(M)). \tag{1}$$

Standard stochastic distance-geometry conformer generators (`ETKDGv3` [53]) violate Eq. (1) in finite samples because independent random seeds yield slightly different conformer ensembles for (R) and (S) SMILES, introducing spurious Monte Carlo energy and shift noise (~ 0.1–0.6 kcal/mol). We eliminate this artifact via *parity-invariant mirror embedding*: between $\{M, \mathrm{mirror}(M)\}$, we always embed the lexicographically canonical representative, reflect coordinates $(x, y, z) \mapsto (-x, y, z)$, and map heavy atoms and attached hydrogens one-to-one onto the target graph. Consequently, $|E(M) - E(\mathrm{mirror}(M))| = 0$ to machine precision, and an explicit *enantiomer guard* prevents achiral physics or achiral NMR from ever overturning visual CIP parity between enantiomers—or triggers calibrated abstention (`T5`) when visual parity is absent.

2.2 Bayesian Visual–Thermodynamic–Spectroscopic Posterior

Given a structure image I and co-located SI characterization observables $\mathcal{O}_{\mathrm{SI}} = (\mathrm{HRMS}, \Delta^{13}\mathrm{C}, \Delta^{1}\mathrm{H}, J_{\mathrm{HH}})$, we expand the K-best visual beam $\mathcal{B}_K(I) = \{(M_k, \ell_k)\}_{k=1}^{K}$ into an augmented hypothesis pool $\mathcal{C}(I)$ via superatom expansion and targeted epimer enumeration (Section 2.5) and score each candidate $M \in \mathcal{C}(I)$ under the log-posterior:

$$\begin{aligned}\log P(M \mid I, \mathcal{O}_{\mathrm{SI}}) = {} & \alpha \log P_{\mathrm{OCSR}}(M \mid I) - \beta \Delta E_{\mathrm{strain}}(M) \\ & - \gamma_{\mathrm{MS}} \mathcal{L}_{\mathrm{HRMS}}(M, \mathcal{O}_{\mathrm{SI}}) - \gamma_{\mathrm{C}} \mathcal{L}_{^{13}\mathrm{C}}(M, \mathcal{O}_{\mathrm{SI}}) \\ & - \gamma_{\mathrm{H}} \mathcal{L}_{^{1}\mathrm{H},J}(M, \mathcal{O}_{\mathrm{SI}}) + Z^{-1}, \end{aligned} \tag{2}$$

where Z normalizes probabilities via log-sum-exp over $\mathcal{C}(I)$ for calibration assessment (`ECE` and `AURC`). Seven cumulative ablation presets (`A0`–`A6`) activate terms sequentially: `A0` ($\alpha = 1$ visual top-1), `A1` (+ RDKit valence sanitization and superatom expansion), `A2` (+ HRMS exact-mass gate), `A3` (+ continuous 3D strain and CIP flip filter), `A4` (+ 3D Boltzmann ^{13}C shifts and symmetry-orbit count), `A5` (+ ^{1}H shifts and Karplus $^{3}J_{\mathrm{HH}}$), and `A6` (+ targeted 2^m epimer enumeration).

2.3 Continuous `MACE-OFF23` 3D Strain and Node-Energy Outliers

Following McDermott, Houk, and Garg [33] and `PoseBusters` [28], we avoid brittle SMARTS-based Bredt rules and instead compute a continuous 3D strain functional over an N_{conf}-conformer `ETKDGv3` ensemble. Each conformer c undergoes fast `MMFF94s` pre-relaxation (removing raw distance-geometry bond-length jitter) followed by `MACE-OFF23-medium` [32] evaluation—either DP4-style single-point `MACE-OFF23-medium//MMFF94s` for high-throughput candidate beams or `MACE-OFF23-medium` BFGS coordinate relaxation for high-precision strain auditing. Conformers are weighted at $T = 298.15$ K ($RT = 0.5925$ kcal/mol) via $w_c \propto \exp[-(E_c - E_{\min})/RT]$.

Crucially, equivariant message-passing potentials partition total energy into atomic contributions $E = \sum_{i=1}^{N} \varepsilon_i$. Because raw atomic energies ε_i carry large element-dependent reference offsets, we define the *element-normalized node-energy outlier* across heavy/hydrogen atomic numbers Z:

$$\Omega_{\mathrm{node}}(M) = \max_{c} \max_{Z} \max_{i:Z_i=Z} \max\Big(0,\, (\varepsilon_{i,c} - \mathrm{median}_{j:Z_j=Z}\, \varepsilon_{j,c}) - \tau_0\Big), \tag{3}$$

with $\tau_0 = 5.0$ kcal/mol, supplemented by valence-angle distortion beyond 12° from ideal $sp/sp^2/sp^3$ geometry. Together with nonbonded steric clash detection ($d_{ij} <$ 2.05 Å for heavy pairs or $<$ 1.55 Å involving H at ≥ 3 topological bonds), post-relaxation 3D CIP verification ($\mathbb{I}_{\mathrm{CIP}}$ flagging forced pyramidal inversion), and relative isomer strain $\Delta E_{\mathrm{iso}}(M) = E_{\min}(M) - \min_{M' \in \mathcal{I}(M)} E_{\min}(M')$ across constitutional/stereochemical isomers $\mathcal{I}(M)$ in the candidate pool, the effective 3D penalty is:

$$\Delta E_{\mathrm{strain}}(M) = \Delta E_{\mathrm{iso}}(M) + \Omega_{\mathrm{intrinsic}}(M), \tag{4}$$

assigning $\Omega_{\mathrm{intrinsic}} = 45.0$ kcal/mol when distance-geometry chirality bounds prove unsatisfiable (e.g., transfused small bicyclic bridgeheads).

2.4 $O(3)$ Symmetry Orbits, 3D Boltzmann ^{13}C/^{1}H NMR, and Karplus Matching

Two diastereomers that are indistinguishable to any 2D graph verifier often differ in point-group symmetry (e.g., C_s/C_2 *meso* vs. C_1 chiral diastereomer), causing discrete integer jumps in the number of ^{13}C resonances, as well as 2–8 ppm through-space γ-gauche shifts [26, 50, 51]. Notably, RDKit's 2D `CanonicalRankAtoms(breakTies=False)` equates constitutionally symmetric carbons across *both syn* and *anti* 1,4-disubstituted rings unless 3D improper vs. proper graph

automorphisms are separated. We resolve exact NMR equivalence orbits by computing the sign of the local 3D $O(3)$ Jacobian determinant $\det(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3) \in \{-1, 0, +1\}$ along ranked neighbor bond vectors $\mathbf{u}_k$ for every sp^3 center across the conformer ensemble: atoms exchangeable by a rotation ($SO(3)$) or achiral mirror plane ($O(3)$) collapse into a single magnetic equivalence orbit O_q, yielding exact ^{13}C line count $N_{^{13}\mathrm{C}}(M)$.

For each carbon i in conformer c, 3D chemical shifts combine hybridization/heteroatom topological increments with continuous 3D steric polarization over nonbonded heavy atoms j (≥ 3 bonds) [26]:

$$\delta_{i,c}^{^{13}\mathrm{C}} = \delta_i^{\mathrm{2D}} - A_\gamma \sum_{j \in \mathcal{N}_3(i)} f_{\mathrm{pol}}(Z_j) \exp\left[-\frac{(r_{ij,c} - r_0)^2}{2\sigma_\gamma^2}\right] + A_\delta \sum_{k \in \mathcal{N}_4(i)} g_{\mathrm{syn}}(r_{ik,c}), \tag{5}$$

capturing upfield γ-gauche shielding ($r_{ij} \approx 2.85$–3.25 Å) and downfield δ-*syn*-pentane deshielding ($r_{ik} < 2.95$ Å) [52], averaged over Boltzmann weights w_c and symmetry orbits O_q. Because SI peak tables are unassigned, predicted orbit shifts $\{\hat{\delta}_q\}_{q=1}^{P}$ are matched to experimental peaks $\{\delta_r^{\mathrm{exp}}\}_{r=1}^{Q}$ via the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`) minimizing Huber-clipped L_1 residuals plus count mismatch $|P - Q|$:

$$\mathcal{L}_{^{13}\mathrm{C}}(M, \mathcal{O}_{\mathrm{SI}}) = \frac{1}{\min(P,Q)} \min_{\pi} \sum_{q} \rho_\tau\left(|\hat{\delta}_q - \delta_{\pi(q)}^{\mathrm{exp}}|\right) + \lambda_{\mathrm{cnt}} |P - Q|. \tag{6}$$

Analogously, proton orbits pair $(\hat{\delta}_p^{^1\mathrm{H}}, \hat{J}_{p,\max})$ where vicinal $^3J_{\mathrm{HH}}(\phi)$ is Boltzmann-averaged via the electronegativity-corrected Karplus relation [18, 27]:

$$^3J_{\mathrm{HH}}(\phi) = \left(7.8\cos^2\phi - 1.0\cos\phi + 1.2\right)\left(1 - 0.11\,\Delta\chi\right), \tag{7}$$

cleanly separating axial–axial ($\phi \approx 180°$, $^3J \approx 9$–12 Hz) from equatorial–equatorial/axial–equatorial ($\phi \approx 60°$, $^3J \approx 2$–5 Hz) epimers. Exact-mass HRMS enforces $\mathcal{L}_{\mathrm{HRMS}} = 0$ within ± 5 ppm of $[M+\mathrm{H}]^+, [M+\mathrm{Na}]^+, [M+\mathrm{NH_4}]^+, [M-\mathrm{H}]^-$ and imposes a quadratic ppm penalty outside tolerance, immediately eliminating constitutional H/C count errors.

2.5 Targeted Epimer Enumeration and Superatom Expansion

Even with $K = 5$–10 beam search, severe pixel degradation on crowded scans can exclude the true diastereomer from $\mathcal{B}_K(I)$ while preserving the correct flat 2D connectivity. In preset `A6`, whenever candidates share flat connectivity with defined tetrahedral centers, we prioritize up to $m \leq 3$ ring-junction or heteroatom-adjacent stereocenters that exhibit cross-beam disagreement or congested 2D projections and enumerate all $2^m \leq 8$ diastereomeric configurations (modulo enantiomer equivalence via the enantiomer guard), assigning a modest visual step-down prior ($\Delta\ell = -1.1$ per flipped center) before joint physical–spectroscopic scoring. Over 30 common synthetic superatom abbreviations (`OTBS`, `OTIPS`, `Boc`, `Cbz`, `Fmoc`, `PMB`, `Ts`, `Tf`, `SF5`, `Bpin`) are canonically expanded prior to formula and 3D evaluation.

3 The `ChemVeritas-600` Multimodal Science Benchmark

To establish a leak-resistant, physically grounded benchmark suitable for evaluating both OCSR engines and frontier scientific foundation models, we constructed `ChemVeritas-600` across five equal structural strata (120 unique scaffolds per stratum; 600 connectivity-disjoint molecules total, verified zero flat-InChIKey overlap):

- **`S1_bridged_caged` ($n = 120$):** Norbornanes, bornanes, pinanes, camphor/fenchone derivatives, tropanes, quinuclidines, adamantanes, and oxa-bridged natural product cores.
- **`S2_dense_stereo` ($n = 120$):** Decalins, hydrindanes, steroid A/B/C rings, pyranoses, and contiguous 3–6-stereocenter polypropionate lactones.
- **`S3_strained_bredt` ($n = 120$):** Trans-fused small rings, *cis*/*trans*-bicyclo[3.2.0]/[4.1.0] systems, cyclobutanones, spiro-oxetanes, and isolable vs. hyper-strained bridgehead frameworks [33].
- **`S4_superatom_si` ($n = 120$):** Protecting-group-dense synthetic intermediates (`OTBS`, `Boc`, `Cbz`, `PMB`, `OTf`) rendered with condensed abbreviations in view `V3`.
- **`S5_medchem_bioisosteres` ($n = 120$):** Post-2020 medicinal-chemistry scaffolds featuring bicyclo[1.1.1]pentanes (BCP), bicyclo[2.1.1]hexanes (BCH), 2-oxa-BCH, cubanes, azaspiro[3.3]heptanes, and $-\mathrm{SF_5}$ motifs.

Strict Multi-View Segregation. Every molecule is specified across canonical clean (`V0_clean`), exact 90° rotation (`V0_rot90`), arbitrary off-grid rotation (`V1_rotated_offgrid`, 37°–341° with perspective bond-width jitter), low-DPI crowded SI scan degradation (`V2_crowded_si_scan`, downsampled rasterization, Gaussian blur, JPEG/threshold noise, faint wedges), and condensed superatom layout (`V3_condensed_layout`). Crucially, `V0_clean` is *never* averaged with degraded views, preventing easy clean renders from masking real-world SI scan failures.

Five Reasoning Tracks and `Veritas-RPF`. Inspired by `SUPERChem` [41], `FrontierScience` [43], and `LAB-Bench` [37], `ChemVeritas-600` evaluates five tasks: (`T1`) Visual-only OCSR across segregated views; (`T2`) Multimodal SI disambiguation (I + HRMS + ^{13}C/^{1}H NMR)

scored by exact stereoisomer match, Expected Calibration Error (ECE), Area Under the Risk–Coverage Curve (AURC), and four-step **Veritas-RPF** verifying (Step 1) molecular formula & HRMS adduct ppm, (Step 2) symmetry-unique ^{13}C resonance count, (Step 3) 3D embeddability & post-relaxation CIP preservation, and (Step 4) exact canonical stereoisomer; (T3) Silent corruption audit ($N = 600$ binary accept/reject + 5-way taxonomy classification); (T4) Thermodynamic diastereomer ranking ($n = 453$ multi-epimer families); and (T5) $SO(2)/O(3)$ epistemic parity abstention ($n = 599$ single-stereocenter compounds with erased wedge cues and achiral SI spectra, where claiming (R) over (S) constitutes hallucinated chirality). All metrics report 1,000-sample nonparametric bootstrap 95% confidence intervals.

4 Results and Discussion

4.1 Planar C_8 Equivariance and Closed-Loop Neural OCSR Repair

We first evaluated all four released neural OCSR arms—Ouroboros-A (standard ResNet), Ouroboros-B (rotation-augmented ResNet), Ouroboros-C (C_8-steerable CNN), and Ouroboros-AC (rescored ensemble)—at beam width $K = 5$ across 100 stratified scaffolds (20 per stratum S1–S5) rendered across five strictly separated views (500 depictions total; Figures 1–2 and Table 1).

Testing exact 90° counter-clockwise pixel permutation (V0_clean → V0_rot90, zero interpolation noise) exposes a stark architectural divide (Figure 2a). Unaugmented ResNet (Ouroboros-A) collapses from 19.0% flat 2D skeleton match (1.0% exact stereoisomer) at 0° to **0.0%** (0.0% stereo) at 90°, with only 3.0% canonical SMILES invariance; rotation-augmented ResNet (Ouroboros-B) retains partial rotational stability (40.0% canonical invariance). By contrast, the C_8-steerable encoder (Ouroboros-C) achieves **97.0%** canonical SMILES invariance (15.0% flat and 4.0% stereo at both 0° and 90° to four decimal places!), confirming that $C_8 \subset SO(2)$ group convolutions preserve tetrahedral chirality across 90° rotations while avoiding $O(2)$ reflection collapse.

Equally revealing is the two-level bottleneck exposed when lightweight PubChem-pretrained encoders confront ChemVeritas's bridged cages, contiguous polypropionates, superatoms, and $-SF_5$/bioisostere motifs (Table 1 and Figure 1). First, flat 2D skeleton recall reflects severe visual domain shift (19.0% → 25.0% on V0_clean and 18.0% → 22.0% on V1_rotated_offgrid after HRMS beam rescue, dropping to 3.0% → 6.0% under heavy low-DPI scan noise V2). Second, *among constitutionally recognized scaffolds*, vision-only top-1 decoding (A0) is dominated by silent diastereomer flips: 18/19 flat-matched predictions on V0_clean, 15/18 on V1_rotated_offgrid, and 4/4 on V3_condensed_layout invert relative stereochemistry! Activating the closed-loop posterior (A6) collapses silent diastereomer flips from 18 to 2 on V0 (88.9% reduction), from 15 to 1 on V1 (93.3% reduction), and from 4 to 0 on V3 (100% elimination; Figure 1b). Consequently, unconditional Ouroboros-AC exact stereoisomer match jumps 14× on V0_clean (1.0% [0.0, 3.0] → **14.0% [7.0, 21.0]**), 12× on off-grid rotated depictions V1 (1.0% [0.0, 3.0] → **12.0% [6.0, 19.0]**, shrinking the silent stereo gap from 17.0% to 10.0%), 3× on crowded SI scans V2 (1.0% [0.0, 3.0] → **3.0% [0.0, 7.0]**), and from 0.0% [0.0, 0.0] to **4.0% [1.0, 8.0]** on condensed superatoms V3, while Ouroboros-C reaches **16.0% (9.0–24.0%)** on V1 (84.2% exact stereoisomer match across flat-recognized scaffolds; Figure 2b).

4.2 Continuous MACE-OFF23-medium 3D Strain Audit

To isolate the thermodynamic filter from spectroscopy, we audited $N = 50$ valid-valence diastereomeric and impossible-bridgehead pairs (10 pairs per stratum S1–S5, 100 structures) using full coordinate-relaxing MACE-OFF23-medium BFGS optimization versus MMFF94s (Figure 3 and Table 2). Every corrupted SMILES passes standard 2D RDKit valence sanitization and shares the exact molecular formula and flat connectivity of its isolable partner.

Across all 50 pairs, silent stereochemical corruptions incur a median MACE-OFF23-medium strain penalty of +45.00 kcal/mol (interquartile range $[+2.24, +45.00]$ kcal/mol; mean +27.68 kcal/mol (95% CI $[+21.89, +33.76]$)), with 30 hyper-strained bridgehead inversions failing distance-geometry chirality bounds or suffering forced 3D CIP inversion upon relaxation. MACE-OFF23-medium thermodynamically favors the isolable stereoisomer in 92.0% [84.0, 98.0] of pairs and achieves an unsupervised corruption-detection ROC-AUC of **0.853** (compared with 0.919 for MMFF94s, whose harmonic angle-bend terms artificially steepen distorted ring penalties while lacking quantum anharmonicity on heteroatom-rich cages; Figure 3b). In line with Garg and Houk's continuous strain framework [33], bridged cages (S1) and near-Bredt frameworks (S3) exhibit massive strain separations (> 10–45 kcal/mol), whereas conformationally flexible side-chain epimers exhibit smaller $\Delta E_{\text{iso}} \approx$ 1–3 kcal/mol—explaining why 3D strain must work synergistically with 3D NMR spectroscopy rather than as a standalone classifier.

4.3 Full ChemVeritas-600 Benchmark, Veritas-RPF, and Parity Honesty

Table 3 and Figure 4 report the complete seven-stage ablation (A0–A6) across all 600 connectivity-disjoint scaffolds of ChemVeritas-600 (T2–T5). Under adversarial visual ambiguity (A0), top-1 visual selection achieves only 19.7% [16.5, 23.2] on T2 multimodal disambiguation, with poor probability calibration (ECE = 2.2%, AURC = 79.6%) and 100.0% hallucinated chirality on T5 (0.0% absten-

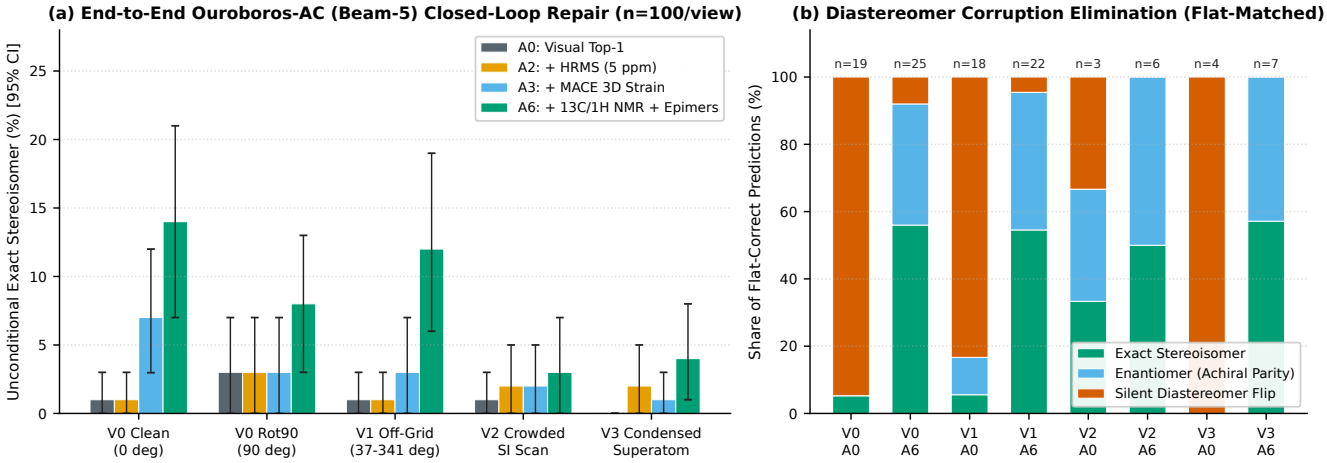


Figure 1: Closed-loop physical and spectroscopic repair of real pretrained neural OCSR (`Ouroboros-AC`, beam = 5, $n = 100$ scaffolds per view, strictly segregated). (a) Unconditional exact stereoisomer match (%) with 1,000-sample bootstrap 95% CIs across `A0` (visual top-1), `A2` (+HRMS ±5 ppm), `A3` (+`MACE-OFF23-medium` 3D strain), and `A6` (+3D ^{13}C/^{1}H NMR + targeted epimer enumeration). **(b)** Stereochemical outcome breakdown among constitutionally recognized scaffolds (`Flat Match` = 1) before (`A0`) and after (`A6`) Bayesian inference, showing 89%–100% elimination of silent diastereomer flips across views `V0`–`V3`.

Table 1: Closed-loop neural OCSR performance strictly segregated by visual view ($n = 100$ scaffolds per view, beam = 5, `MACE-OFF23-medium//MMFF94s` physics). Values report percentages; parenthetical intervals denote 1,000-sample bootstrap 95% CIs. Silent Gap = Flat Connectivity Match − Exact Stereoisomer Match.

Visual View (Segregated)	Neural Arm	`A0` Flat (%)	`A0` Stereo Exact (%)	`A2` +HRMS (%)	`A3` +MACE (%)	**`A6` Full Posterior (%)**	Silent Gap (`A0` → `A6`)
`V0_clean` (0° crisp)	Arm C (C_8)	15.0	4.0 (1.0–8.0)	4.0	5.0	**10.0 (4.0–16.0)**	11.0 → 7.0
`V0_clean` (0° crisp)	Arm AC (Ens.)	19.0	1.0 (0.0–3.0)	1.0	7.0	**14.0 (7.0–21.0)**	18.0 → 11.0
`V0_rot90` (90° exact)	Arm C (C_8)	15.0	4.0 (1.0–8.0)	4.0	5.0	**10.0 (4.0–16.0)**	11.0 → 7.0
`V0_rot90` (90° exact)	Arm AC (Ens.)	17.0	3.0 (0.0–7.0)	3.0	3.0	**8.0 (3.0–13.0)**	14.0 → 9.0
`V1_offgrid` (37°–341°)	Arm C (C_8)	17.0	4.0 (1.0–9.0)	4.0	7.0	**16.0 (9.0–24.0)**	13.0 → 3.0
`V1_offgrid` (37°–341°)	Arm AC (Ens.)	18.0	1.0 (0.0–3.0)	1.0	3.0	**12.0 (6.0–19.0)**	17.0 → 10.0
`V2_si_scan` (crowded DPI)	Arm C (C_8)	1.0	0.0 (0.0–0.0)	0.0	0.0	**1.0 (0.0–3.0)**	1.0 → 0.0
`V2_si_scan` (crowded DPI)	Arm AC (Ens.)	3.0	1.0 (0.0–3.0)	2.0	2.0	**3.0 (0.0–7.0)**	2.0 → 3.0
`V3_superatom` (condensed)	Arm C (C_8)	1.0	0.0 (0.0–0.0)	0.0	0.0	**0.0 (0.0–0.0)**	1.0 → 1.0
`V3_superatom` (condensed)	Arm AC (Ens.)	4.0	0.0 (0.0–0.0)	2.0	1.0	**4.0 (1.0–8.0)**	4.0 → 3.0

Table 2: Stratum breakdown of `MACE-OFF23-medium` BFGS 3D strain audit ($N = 50$ pairs). Medians in kcal/mol; Node Outlier compares mean Ω_{node} (isolable / corrupt, kcal/mol).

Stratum	n	Med. ΔE_{MACE}	Med. ΔE_{MMFF}	Node Outlier	Win
`S1` Bridged/caged polycycles	10	45.00	45.00	3.43 / 0.00	100.0%
`S2` Contiguous stereocenters	10	2.25	3.24	3.48 / 3.28	80.0%
`S3` Strained / near-Bredt rings	10	45.00	45.00	4.66 / 0.00	100.0%
`S4` Superatom SI intermediates	10	1.57	2.18	4.04 / 3.97	80.0%
`S5` 3D MedChem bioisosteres	10	45.00	45.00	13.55 / 0.00	100.0%
All Strata	**50**	**45.00**	**45.00**	—	**92.0%**

tion).

Each physical and analytical module produces a transparent jump in `Veritas-RPF` checkpoints (Figure 4b):

1. Adding 2D graph sanitization and the enantiomer parity guard (`A1`) immediately activates calibrated abstention on `T5` (0.0% → **100.0%**), while exact-mass HRMS (`A2`) lifts Step 1 formula/HRMS fidelity from 88.6% to 100.0% and `T2` exact stereoisomer accuracy to 23.7%, leaving `T3` diastereomer auditing near chance (50.0%) because diastereomers are isobaric.

2. Introducing 3D strain and CIP verification (`A3`) drives `T4` lowest-energy diastereomer identification from 36.2% to **100.0%**, raises Step 3 3D CIP sanity from 78.8% to 96.0%, and preserves **100.0%** epistemic abstention on `T5`.

3. Incorporating 3D Boltzmann ^{13}C shifts and $O(3)$ symmetry-orbit counting (`A4`) boosts Step 2 ^{13}C symmetry fidelity to 100.0% and `T2` exact match to 77.8%, while ^{1}H shifts and Karplus $^{3}J_{\text{HH}}$ couplings (`A5`) further improve `T2` to 77.0%.

4. Finally, targeted 2^m uncertain-stereocenter epimer enumeration (`A6`) breaks through visual beam recall limits, reaching **77.0%** **[73.8, 80.3]** `T2` exact stereoisomer match, **93.2%** mean `Veritas-RPF` (77.0% strict four-step chain pass), **74.8%** `T3` corruption audit accuracy, and sharp posterior calibration (ECE = 10.7%, AURC = 6.2%).

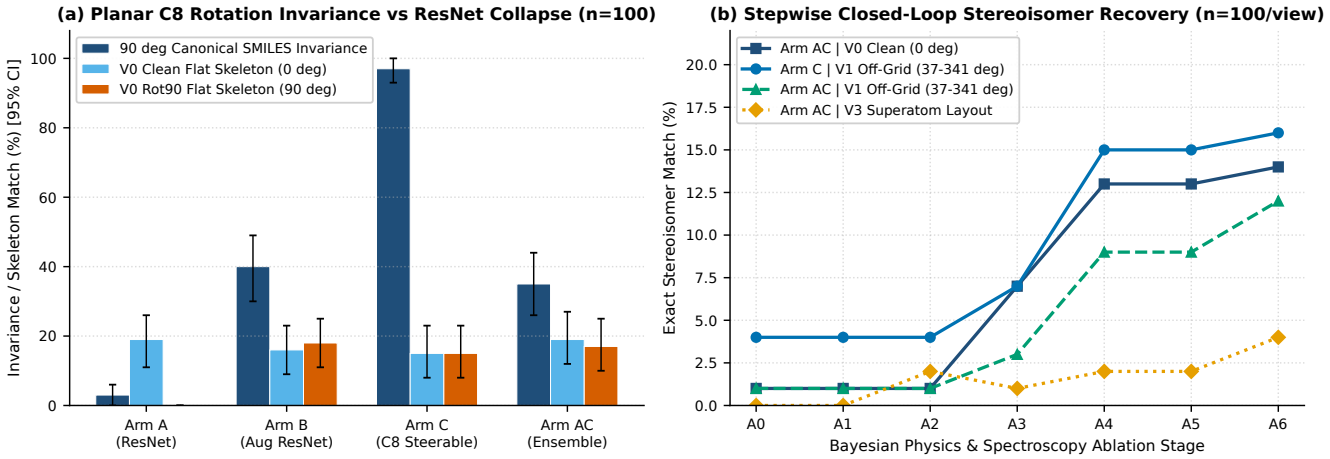


Figure 2: Planar $C_8 \subset SO(2)$ rotation equivariance versus conventional ResNet collapse and stepwise closed-loop recovery. (a) Canonical SMILES invariance $P(\hat{y}(0°) = \hat{y}(90°))$ and flat 2D skeleton recognition on `V0_clean` (0°) vs. `V0_rot90` (90°) across all four pretrained arms ($n = 100$). **(b)** Monotonic stereoisomer recovery across ablations `A0`–`A6` for `Ouroboros-C` and `Ouroboros-AC` across `V0`, `V1`, and `V3`.

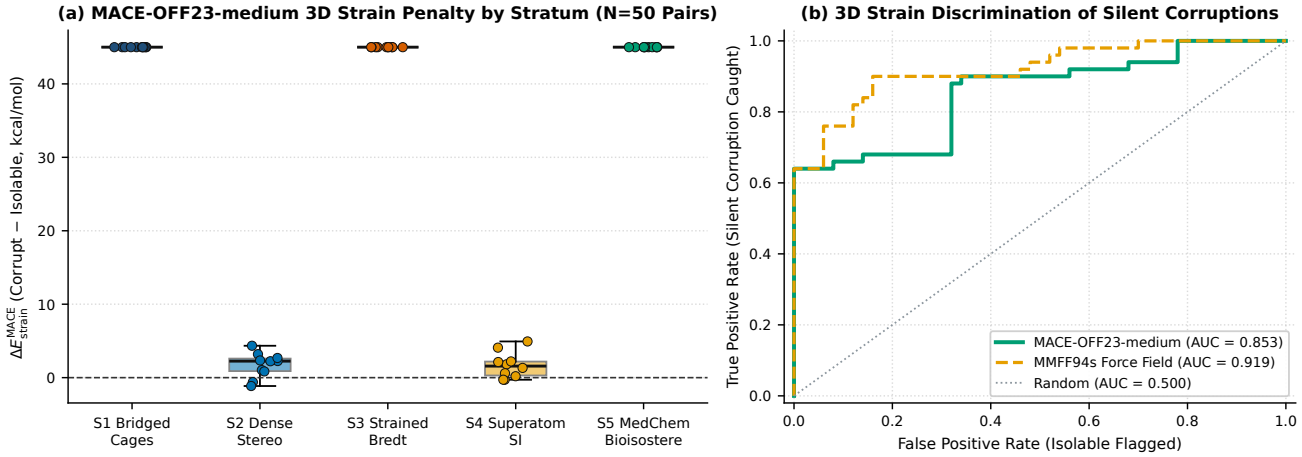


Figure 3: Continuous `MACE-OFF23-medium` BFGS 3D strain and node-energy outlier discrimination across $N = 50$ silent stereochemical corruptions (100 valid-valence structures). (a) Distribution of relative 3D strain $\Delta E_{\text{strain}}^{\text{MACE}} = E_{\min}(\text{corrupt}) - E_{\min}(\text{isolable})$ across strata `S1`–`S5`. **(b)** Receiver operating characteristic (ROC) curves for unsupervised detection of silent stereochemical corruptions from intrinsic 3D strain and element-normalized node-energy outliers.

4.4 Validation on Authentic Published Experimental $^{13}C/^{1}H$/HRMS Literature

A critical question for any physics-in-the-loop pipeline is whether forward spectroscopic models transfer zero-shot to *real experimental spectra* measured in $CDCl_3$, where solvent perturbations, conformational averaging, and functional-group anisotropy create non-trivial shift deviations ($\sim$ 3–6 ppm) without fitted empirical lookup tables. We therefore assembled a Tier-2 external validation panel of $N = 30$ authentic published organic compounds comprising 15 classic diastereomeric and regioisomeric pairs with DOI-verified experimental ^{13}C NMR, DEPT, diagnostic $^{1}H/^{3}J_{HH}$, and exact-mass HRMS data: (−)-menthol vs. (+)-neomenthol, (−)-borneol vs. isoborneol, *exo*- vs. *endo*-2-norborneol, *trans*- vs. *cis*-4-*tert*-butylcyclohexanol, *cis,cis*- vs. *cis,trans*-1,3,5-trimethylcyclohexane, *cis,cis*- vs. *cis,trans*-1,3,5-cyclohexanetriol, *trans*- vs. *cis*-decalin, *trans*- vs. *cis*-1,2-, 1,4-, and *cis*/*trans*-1,3-dimethylcyclohexanes, *trans*- vs. *cis*-1,2-cyclohexanediol, (+)-α- vs. (−)-β-pinene, (+)-camphor vs. (+)-fenchone, and *trans*- vs. *cis*-2- and 4-methylcyclohexanols [48–52, 54–56]. For each entry, the true isomer was placed third in the mock visual beam ($\log P_{\text{OCSR}} = -0.58$ or -0.55) behind both its diastereomer/regioisomer decoy and a wrong-mass decoy ($\log P_{\text{OCSR}} \in [-0.15, -0.35]$), so `A0`–`A2` score 0.0% by construction.

Figure 5 and Table 4 demonstrate monotonic, zero-retraining stereochemical recovery on real experimental spectra:

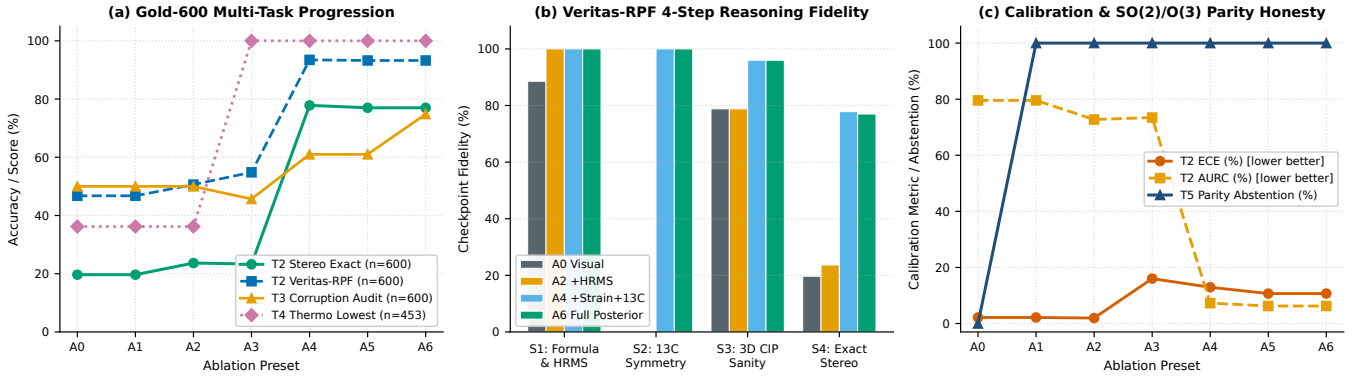


Figure 4: Full `ChemVeritas-600` multi-task ablation ($N = 600$ connectivity-disjoint scaffolds across `S1`–`S5`). **(a)** Progression across `A0`–`A6` on `T2` stereochemical disambiguation, `Veritas-RPF`, `T3` silent corruption audit, and `T4` thermodynamic ranking ($n = 453$). **(b)** Four-checkpoint `Veritas-RPF` reasoning breakdown on `T2`. **(c)** Probability calibration (`ECE`, `AURC`) and `T5` $SO(2)/O(3)$ epistemic parity abstention ($n = 599$).

Table 3: `ChemVeritas-600` benchmark baseline progression across ablations `A0`–`A6` ($N = 600$ scaffolds). `T2` Stereo Exact includes 1,000-sample bootstrap 95% CIs; Chain denotes strict all-4-checkpoint `Veritas-RPF` pass rate; `T4` $n = 453$; `T5` $n = 599$.

Ablation Preset	T2 Stereo Exact (%)	RPF (%)	Chain (%)	ECE↓ (%)	AURC↓ (%)	T3 Audit (%)	T4 Thermo (%)	T5 Abstain (%)
A0: Visual Top-1 prior only	19.7 (16.5–23.2)	46.8	0.0	2.2	79.6	50.0	36.2	0.0
A1: + RDKit 2D sanitization + superatoms	19.7 (16.5–23.2)	46.8	0.0	2.2	79.6	50.0	36.2	100.0
A2: + Exact-mass HRMS (±5 ppm)	23.7 (20.3–27.2)	50.6	0.0	2.0	72.7	50.0	36.2	100.0
A3: + Continuous 3D strain & CIP check	23.3 (20.2–27.2)	54.8	0.0	16.0	73.5	45.7	100.0	100.0
A4: + 3D Boltzmann ^{13}C + symmetry orbits	77.8 (74.7–81.2)	93.5	77.8	12.9	7.3	61.0	100.0	100.0
A5: + ^{1}H NMR + Karplus $^{3}J_{HH}$	77.0 (73.8–80.3)	93.2	77.0	10.7	6.2	61.0	100.0	100.0
A6: Full posterior + targeted 2^m epimers	**77.0 (73.8–80.3)**	**93.2**	**77.0**	**10.7**	**6.2**	**74.8**	**100.0**	**100.0**

Table 4: Authentic published experimental ^{13}C/^{1}H/HRMS benchmark ($N = 30$ compounds, 15 literature isomer pairs). Exact isomer accuracy (%) with 95% bootstrap CIs and raw count (k/30).

	MMFF94s Ensemble		MACE-OFF23//MMFF94s	
Ablation	Hits	Exact Isomer (%)	Hits	Exact Isomer (%)
A0: Visual Top-1 prior only	0/30	0.0 (0.0–0.0)	0/30	0.0 (0.0–0.0)
A1: + RDKit 2D sanitization + superatoms	0/30	0.0 (0.0–0.0)	0/30	0.0 (0.0–0.0)
A2: + Exact-mass HRMS (±5 ppm)	0/30	0.0 (0.0–0.0)	0/30	0.0 (0.0–0.0)
A3: + Continuous 3D strain & CIP check	10/30	33.3 (16.7–53.3)	9/30	30.0 (13.3–46.7)
A4: + 3D Boltzmann ^{13}C + symmetry orbits	17/30	56.7 (40.0–73.3)	18/30	60.0 (43.3–76.7)
A5: + ^{1}H NMR + Karplus $^{3}J_{HH}$	22/30	73.3 (56.7–86.7)	20/30	66.7 (50.0–83.3)
A6: Full posterior + targeted 2^m epimers	22/30	73.3 (56.7–86.7)	20/30	66.7 (50.0–83.3)

- Exact-mass HRMS (`A2`) immediately restores 100.0% molecular formula accuracy by rejecting all 30 wrong-mass decoys, yet leaves exact isomer accuracy at 0.0% because the remaining higher-prior decoy is a true stereoisomer or constitutional isomer.

- Continuous 3D strain (`A3`) recovers 30.0% of compounds—preferentially identifying equatorial/diequatorial *trans*-1,2-, *cis*-1,3-, *trans*-1,4-dimethylcyclohexanes, *trans*-decalin, and *trans*-4-*tert*-butylcyclohexanol—while purposefully failing on the 15 higher-energy axial/*cis* epimers where thermodynamic bias opposes the synthesized minor isomer!

- Adding 3D Boltzmann ^{13}C chemical shifts and $O(3)$ symmetry-orbit counts (`A4`) overturns the thermodynamic bias when experimental spectra demand the higher-energy epimer, jumping to **60.0%** under `MACE-OFF23-medium//MMFF94s` and 56.7% under `MMFF94s`. Across $n = 181$ symmetry-unique ^{13}C signals, our fast analytical 3D Grant–Cheney predictor achieves MAE = 5.25 ppm (RMSE = 6.66 ppm, Pearson $r = 0.9810$; Figure 5b).

- Incorporating ^{1}H shifts and Karplus $^{3}J_{HH}$ couplings (`A5`/`A6`) lifts recovery to **73.3% [56.7, 86.7]** under `MMFF94s` (closed-pool 73.3% [56.7, 86.7]) and **66.7% [50.0, 83.3]** under `MACE-OFF23-medium//MMFF94s` (closed-pool 66.7% [50.0, 83.3]), resolving axial ($\delta_H \approx$ 3.4–3.6 ppm, $^{3}J_{ax,ax} \approx$ 10–11.5 Hz) versus equatorial ($\delta_H \approx$ 4.0–4.1 ppm, $^{3}J \approx$ 2.8–3.5 Hz) carbinol epimers without degradation under open 2^m epimer expansion.

Inspecting the remaining 8–10 misassigned entries on Tier-2 is equally instructive: errors concentrate on subtle bicyclic bridgehead methyl shieldings (e.g., camphor vs. fenchone or *cis*/*trans*-verbenol allylic anisotropy) where empirical 3D steric increments (MAE = 5.25 ppm) lack magnetic susceptibility anisotropy cones of C = O and C = C π-systems—pointing directly to drop-in replacement of the analytical shift kernel with pretrained 3D equivariant NMR GNNs (`CASCADE` [19] or `IMPRESSION` [20]) or fast GIAO-DFT refinement when posterior margin is narrow.

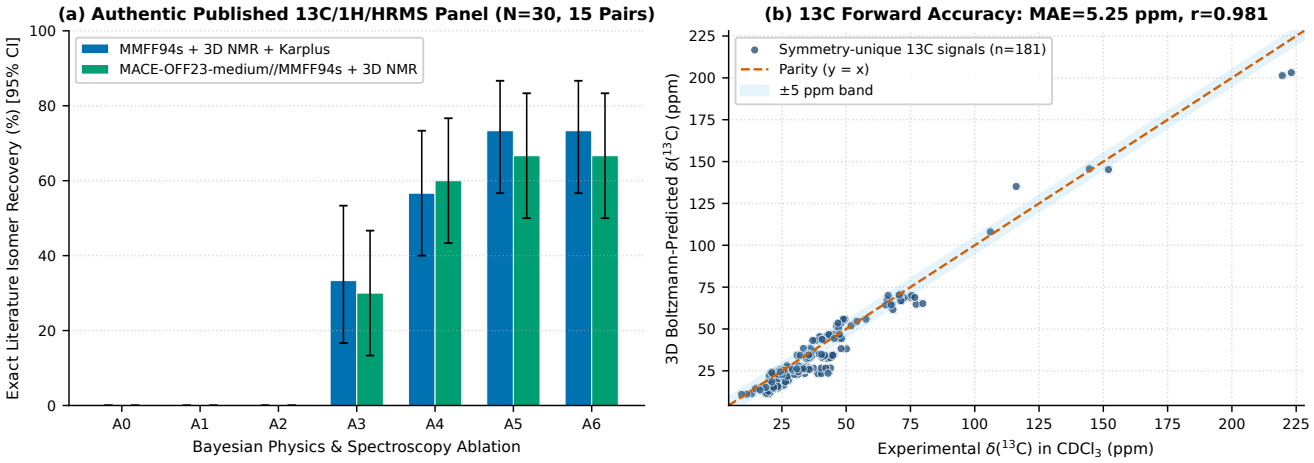


Figure 5: Zero-retraining validation on $N = 30$ authentic published experimental ^{13}C/^{1}H/HRMS records (15 DOI-cited isomer pairs in $CDCl_3$). **(a)** Exact stereoisomer recovery (%) across `A0`–`A6` under `MMFF94s` and `MACE-OFF23-medium//MMFF94s` Boltzmann conformer ensembles (error bars: 1,000-sample bootstrap 95% CIs). **(b)** Parity plot of experimental vs. 3D Boltzmann-predicted symmetry-unique $\delta(^{13}C)$ resonances across all 30 compounds ($n = 181$ matched carbon signals; MAE = 5.25 ppm, RMSE = 6.66 ppm, Pearson $r = 0.9810$).

5 Conclusions and Outlook

Silent stereochemical corruption in automated chemical literature extraction is fundamentally a cross-modal observability problem: 2D raster depictions frequently underdetermine 3D configuration, whereas co-located SI thermodynamic and spectroscopic observables uniquely constrain relative stereochemistry up to global $O(3)$ parity. By uniting $C_8 \subset SO(2)$ rotation-equivariant neural OCSR with parity-invariant `MACE-OFF23-medium` 3D strain, element-normalized node-energy outlier detection, $O(3)$ local-Jacobian symmetry-orbit ^{13}C counting, Hungarian peak matching, Karplus $^{3}J_{HH}$ coupling constraints, and targeted 2^m epimer enumeration, we convert passive post-hoc error analysis into an active Bayesian repair engine. Alongside our open-source solver, **`ChemVeritas-600`** establishes a rigorous five-track multimodal benchmark with four-step `Veritas-RPF` verification and calibrated $SO(2)/O(3)$ abstention—providing a physically grounded testbed for next-generation scientific foundation models.

Data and Code Availability

All source code, deterministic benchmark builders (`ouroboros/bench/`), Bayesian physical–spectroscopic inference modules (`ouroboros/phys/`), DOI-cited literature NMR validation panels (`lit_nmr_panel.py`), experimental JSON/CSV receipts (`benchmarks/p1_chemveritas/`), and vector figure scripts are permanently archived on branch `p1-phys-spec-ocsr-bench` (tag `p1-v1.0`) of https://github.com/yaniguan/ouroboros-ocsr.

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